Staggered 3-mesh method for CCD correlation energy calculation

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Using an arbitrary uniform mesh $\mathcal{K}$, we first construct three staggered meshes $\mathcal{K}_0, \mathcal{K}_{\frac{1}{3}}, \mathcal{K}_{\frac{2}{3}}$ by shifting $\mathcal{K}$ by 0-, $\frac{1}{3}$- and $\frac{2}{3}$-mesh-sizes in all periodic directions. For brevity, we use capital letters to represent the band index and momentum vector pair, e.g., $P \leftrightarrow (p, \mathbf{k}_p)$, and we use $P = x \in \{0, \frac{1}{3}, \frac{2}{3}\}$ to indicate that momentum vector $\mathbf{k}_p$ is sampled in $\mathcal{K}_x$. A simple rule is that:

For $\mathbf{k}_1 \in x$ and $\mathbf{k}_2 \in y$ with $x \neq y \in \{0, \frac{1}{3}, \frac{2}{3}\}$, $\mathbf{k}_1 + \mathbf{k}_2 \in (x + y) = 2z$ with $z = \{0, \frac{1}{3}, \frac{2}{3}\} \setminus \{x, y\}$.

By momentum conservation, the double amplitude t_{IJ}^{AB} in CCD (or MP3) can have momentum vectors $(\mathbf{k}_i, \mathbf{k}_j, \mathbf{k}_a, \mathbf{k}_b)$ sampled in the following three forms:

- t_{xx}^{yz} with $\{x, y, z\} = \{0, \frac{1}{3}, \frac{2}{3}\}$, i.e., I and J are on mesh x, A and B are on the other two y and z.
- t_{xy}^{yx} with $\{x, y\} \subset \{0, \frac{1}{3}, \frac{2}{3}\}$.
- t_{xy}^{zz} with $\{x, y, z\} = \{0, \frac{1}{3}, \frac{2}{3}\}$.

Each of the three forms have exactly 6 instances and 3 if taking $t_{IJ}^{AB} = t_{JI}^{BA}$ symmetry into account.

0.1 3-mesh method for energy calculation

The CCD correlation energy is computed as

$$E_{\text{corr}} = \frac{1}{N_{\mathbf{k}}} \sum_{IJAB} \left(2 \langle IJ|AB\rangle - \langle IJ|BA\rangle\right) t_{IJ}^{AB}.$$

Following the idea of staggered mesh method, I should not be sampled on the same mesh as A or B due to the ERIs and the double amplitude. There are 12 possible ways of mesh sampling for energy calculation:

- $t_{IJ}^{AB} = t_{xx}^{yz}$
$$E_{\text{corr}} = \frac{1}{N_{\mathbf{k}}} \sum_{IJAB} \left(2 \langle xx|yz\rangle - \langle xx|zy\rangle\right) t_{xx}^{yz}.$$
- $t_{IJ}^{AB} = t_{xy}^{zz}$
$$E_{\text{corr}} = \frac{1}{N_{\mathbf{k}}} \sum_{IJAB} \left(2 \langle xy|zz\rangle - \langle xy|zz\rangle\right) t_{xy}^{zz}.$$

0.2 Amplitude equation

Assuming a diagonal Fock matrix from the Hartree-Fock calculation, the amplitude equation for CCD double amplitudes writes as

$$\varepsilon_{IJAB} t_{IJ}^{AB} = \langle AB|IJ\rangle + \sum_{KL} \chi_{IJ}^{KL} t_{KL}^{AB} + \sum_{CD} \chi_{CD}^{AB} t_{IJ}^{CD} + \mathcal{P} \sum_{C} \kappa_C^A t_{IJ}^{CB} - \mathcal{P} \sum_{K} \kappa_I^K t_{KJ}^{AB} + \mathcal{P} \sum_{KC} (2\chi_{IC}^{AK} - \chi_{CI}^{AK}) t_{KJ}^{CB} - \mathcal{P} \sum_{KC} \chi_{IC}^{AK} t_{KJ}^{BC} - \mathcal{P} \sum_{KC} \chi_{CI}^{BK} t_{KJ}^{AC},$$

which corresponds to the $1, 2, 4, 5, 6, 10, 11$, and 12th terms in [1]. The 4-index intermediate blocks above are defined as

- $\langle oo|oo\rangle$ intermediate block: χ_{IJ}^{KL}

$$\chi_{IJ}^{KL} = \langle KL|IJ\rangle + \sum_{CD} \langle KL|CD\rangle \, t_{IJ}^{CD},$$

- $\langle vv|vv\rangle$ intermediate block: χ_{CD}^{AB}

$$\chi_{CD}^{AB} = \langle AB|CD\rangle \,,$$

- $\langle vo|ov\rangle$ intermediate block: χ_{IC}^{AK} (including χ_{CI}^{BK})

$$\chi_{IC}^{AK} = \langle AK|IC\rangle - \frac{1}{2} \sum_{LD} \langle LK|DC\rangle \, t_{IL}^{DA} + \frac{1}{2} \sum_{LD} \left(2 \langle LK|DC\rangle - \langle LK|CD\rangle\right) t_{IL}^{AD},$$

- $\langle vo|vo\rangle$ intermediate block: χ_{CI}^{AK}

$$\chi_{CI}^{AK} = \langle AK|CI\rangle - \frac{1}{2} \sum_{LD} \langle LK|CD\rangle \, t_{IL}^{DA}.$$

All these blocks, say χ_{RS}^{PQ}, satisfy the momentum conservation

$$\mathbf{k}_p + \mathbf{k}_q = \mathbf{k}_r + \mathbf{k}_s + \mathbb{L}^*$$

and have singularity at "zero momentum transfer" points $\mathbf{k}_p - \mathbf{k}_r \in \mathbb{L}^*$. The 2-index intermediate blocks (excluding the orbital energies in [1]) in the equation are defined as

$$\kappa_C^A = -\sum_{KLD} \left(2 \langle KL|CD\rangle - \langle KL|DC\rangle\right) t_{KL}^{AD},$$
$$\kappa_I^K = \sum_{LCD} \left(2 \langle KL|CD\rangle - \langle KL|DC\rangle\right) t_{IL}^{CD},$$

where the momentum conservation requires that $\mathbf{k}_a = \mathbf{k}_c$ and $\mathbf{k}_k = \mathbf{k}_i$, respectively.

0.3 3-mesh method for amplitude equation

In the following discussion, we separate the 3-mesh calculation for the amplitude equation and the intermediate blocks. Given a staggered mesh sampling for t_{IJ}^{AB} with $I \neq A$, the 3-mesh method for each term in the amplitude equation of t_{IJ}^{AB} have the constraints listed as follows: (define $S_0 = \{0, \frac{1}{3}, \frac{2}{3}\}$)

- 2nd term: $\sum_{KL} \chi_{IJ}^{KL} t_{KL}^{AB}$
  Constraints: $K + L = A + B = I + J, K \neq I, K \neq A$
  $$K = I + \Delta,\ L = J - \Delta, \quad \Delta \in S_0 \setminus \{0, A - I\}$$

- 3rd term: $\sum_{CD} \chi_{CD}^{AB} t_{IJ}^{CD}$
  Constraints: $C + D = A + B, C \neq A, C \neq I$
  $$C = I + \Delta,\ D = J - \Delta, \quad \Delta \in S_0 \setminus \{0, A - I\}$$

- (6+7)th term: $\sum_{KC} (2\chi_{IC}^{AK} - \chi_{CI}^{AK}) t_{KJ}^{CB} - \chi_{IC}^{AK} t_{KJ}^{BC}$
  Constraints: $K - C = I - A \neq \mathbf{0}, K \neq B, K \neq I$
  $$K = I + \Delta,\ C = A + \Delta, \quad \Delta \in S_0 \setminus \{0, B - I\}$$

- 8th term: $\sum_{KC} \chi_{CI}^{BK} t_{KJ}^{AC}$
  Constraints: $K - C = \boldsymbol{I - B}, K \neq I, K \neq A$
  $$K = I + \Delta,\ C = B + \Delta, \quad \Delta \in S_0 \setminus \{0, A - I\}$$

Note that we combine the 6th and 7th terms together due to the notable connections among the involved terms. For the three types of mesh sampling for t_{IJ}^{AB}, the above constraints can determine the sampling mesh for those intermediate momentum vectors, i.e., K, L, C, D, as

t_{IJ}^{AB}	t_{xx}^{yz}	t_{yz}^{xx}	t_{xy}^{yx}
2nd term (K, L)	(z, y)	(z, y)	(z, z)
3rd term (C, D)	(z, y)	(z, y)	(z, z)
(6+7)th term (K, C)	(y, z)	(z, y)	(y, z) or (z, x)
8th term (K, C)	(z, y)	(z, y)	(z, z)

Note that there are two possible choices for (6+7)th terms for $t_{IJ}^{AB} = t_{xy}^{yx}$ and further investigation may be needed (**UNDONE**). For the 6-8th terms, the mesh sampling for intermediate momentum vectors for the symmetrized terms (i.e. when $\mathcal{P}$ switches (I, A) and (J, B)) can be similarly derived.

0.4 3-mesh method for 4-index intermediate blocks

- $\langle oo|oo\rangle$-$\chi_{IJ}^{KL} = \langle KL|IJ\rangle + \sum_{CD} \langle KL|CD\rangle\, t_{IJ}^{CD}$
  Condition from amplitude equation: $K + L = I + J, K \neq I$
  Constraints: $C + D = I + J, C \neq K, C \neq I$
  $$C = I + \Delta,\ D = J - \Delta, \quad \Delta \in S_0 \setminus \{0, K - I\}$$

- $\langle vv|vv\rangle$-$\chi^{AB}_{CD} = \langle AB|CD\rangle$
- $\langle vo|ov\rangle$-$\chi^{AK}_{IC} = \langle AK|IC\rangle - \frac{1}{2}\sum_{LD}\langle LK|DC\rangle\, t^{DA}_{IL} + \frac{1}{2}\sum_{LD}\left(2\langle LK|DC\rangle - \langle LK|CD\rangle\right) t^{AD}_{IL}$
  Condition from amplitude equation: $A + K = I + C, A \neq I, K \neq I$
  Constraints: $L - D = C - K = A - I \neq 0, D \neq I, D \neq K$

$$D = I + \Delta,\ L = A + \Delta, \quad \Delta \in S_0 \setminus \{0, K - I\}$$

- $\langle vo|vo\rangle$-$\chi^{AK}_{CI} = \langle AK|CI\rangle - \frac{1}{2}\sum_{LD}\langle LK|CD\rangle\, t^{DA}_{IL}$
  Condition from amplitude equation: $A + K = C + I, A \neq C$
  Constraints: $L - D = C - K = A - I \neq 0, D \neq K, D \neq I$

$$D = I + \Delta,\ L = A + \Delta, \quad \Delta \in S_0 \setminus \{0, K - I\}$$

For these 4-index blocks, the mesh sampling of their momentum vectors also have at most three forms, i.e., χ^{yz}_{xx}, χ^{xx}_{yz} and χ^{yx}_{xy}. In each of these forms, the corresponding mesh sampling for the intermediate momentum vectors can be derived as follows:

χ^{PQ}_{RS}	χ^{yz}_{xx}	χ^{xx}_{yz}	χ^{yx}_{xy}
$\chi^{KL}_{IJ} \leftarrow (C, D)$	(z, y)	(z, y)	(z, z)
χ^{AB}_{CD}	-	-	-
$\chi^{AK}_{IC} \leftarrow (L, D)$	(z, y)	(z, y)	-
$\chi^{AK}_{CI} \leftarrow (L, D)$	(z, y)	(z, y)	(z, z)

Note that in the amplitude equation, there is no χ^{AK}_{IC} sampled in the form χ^{yx}_{xy} due to the constraint that $K \neq I$ for the calculation of (6+7)th terms.

0.5 3-mesh method for 2-index intermediate blocks

- virtual-$\kappa^{A}_{C} = \Big[\langle KL|CD\rangle + \langle KL|DC\rangle\Big] t^{AD}_{KL}$
  Constraints 1: $A = C$
  Constraints 2: $L - D = C - K, K \neq C, K \neq D$

$$K \in S_0 \setminus \{A\},\ L \in S_0 \setminus \{A\},\ D = K + L - A$$

- occupied-$\kappa^{K}_{I} = \Big[\langle KL|CD\rangle + \langle KL|DC\rangle\Big] t^{CD}_{IL}$
  Constraints 1: $K = I$
  Constraints 2: $L - D = C - K, C \neq K, C \neq L$

$$C \in S_0 \setminus \{I\}, D \in S_0 \setminus \{I\},\ L = C + D - I$$

For both 2-index intermediate blocks, there are 4 admissible choices of mesh sampling of those intermediate momentum vectors as follows

κ_P^Q	κ_x^x
$\kappa_C^A \leftarrow (K, L, D)$	$(x+\frac{1}{3}, x+\frac{1}{3}, x+\frac{2}{3})$, $(x+\frac{1}{3}, x-\frac{1}{3}, x)$, $(x-\frac{1}{3}, x+\frac{1}{3}, x)$, $(x-\frac{1}{3}, x-\frac{1}{3}, x-\frac{2}{3})$
$\kappa_I^K \leftarrow (C, D, L)$	$(x+\frac{1}{3}, x+\frac{1}{3}, x+\frac{2}{3})$, $(x+\frac{1}{3}, x-\frac{1}{3}, x)$, $(x-\frac{1}{3}, x+\frac{1}{3}, x)$, $(x-\frac{1}{3}, x-\frac{1}{3}, x-\frac{2}{3})$

As a result, it may also be worthy of further investigation about which mesh sampling to use in practice (**UNDONE**).

References

[1] S. Hirata, R. Podeszwa, M. Tobita, and R. J. Bartlett. Coupled-cluster singles and doubles for extended systems. *The Journal of chemical physics*, 120(6):2581–2592, 2004.